

AWS IAM Inline Policies

To Begin with the Lab

Summary of the Lab

In this lab, **AWS IAM Inline Policies** were explored as a policy type that is directly embedded into a single IAM user, group, or role. Two normal IAM users were first created to show how a reusable policy can be attached to multiple entities, while a separate inline policy user was created to demonstrate the one-to-one relationship of inline policies. A custom JSON policy was written to provide specific permissions, and then it was attached directly to the selected user instead of being created as a reusable standalone policy. The lab showed that when the IAM entity is deleted, the inline policy is deleted with it, unlike managed policies that remain available for reuse. Finally, the lesson explained why inline policies are useful for temporary, highly specific, or compliance-driven access requirements.

Understanding IAM Inline Policies

An **IAM Inline Policy** is a policy that is directly attached to one IAM user, one IAM group, or one IAM role. It exists so a permission rule can be tightly coupled to a single identity without being reusable by other identities. This makes inline policies useful when access must be temporary, highly specific, or not shared across multiple users. Unlike managed policies, inline policies do not have their own ARN and are deleted automatically when the attached IAM entity is deleted.

Example:

A temporary project user needs read-only access to **Amazon EC2** in the ap-south-1 Region only. A custom JSON inline policy is attached directly to that single IAM user, and no other user can reuse it. When the user is deleted after the project ends, the inline policy disappears automatically as well.

Lab Steps

Step 1 – Create the Sample IAM Users

- Sign in to the AWS Management Console.
- Open **IAM** → **Users**.
- Click **Create user**.

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IAM users (3) [Info](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

☐	User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID
☐	[redacted]	/	0	7 hours ago	-	8 days	7 hours ago	[redacted]
☐	[redacted]	/	0	6 hours ago	-	7 hours	6 hours ago	-
☐	[redacted]	/	0	1 hour ago	-	18 days	1 hour ago	-

[Delete](#) [Create user](#)

- Create **user-1**.
- Enable console access.
- Set a password.
- Leave the user without permissions.

User details

User name
user-1

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , @ _ - (hyphen)

☒ Provide user access to the AWS Management Console - optional

Are you providing console access to a person?

User type

☐ Specify a user in Identity Center - Recommended

We recommend that you use Identity Center to provide console access to a person. With Identity Center, you can centrally manage user access to their AWS accounts and cloud applications.

☒ I want to create an IAM user

Console password

☐ Autogenerated password

You can view the password after you create the user.

☒ Custom password

Enter a custom password for the user.

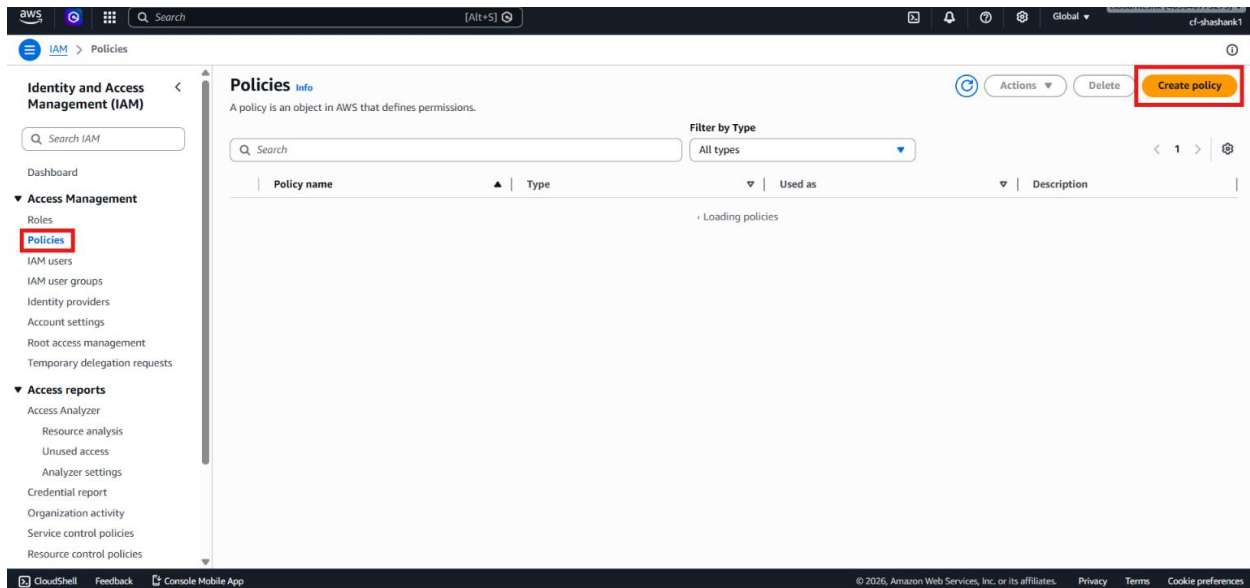
☐ Show password

☐ Users must create a new password at next sign-in - Recommended

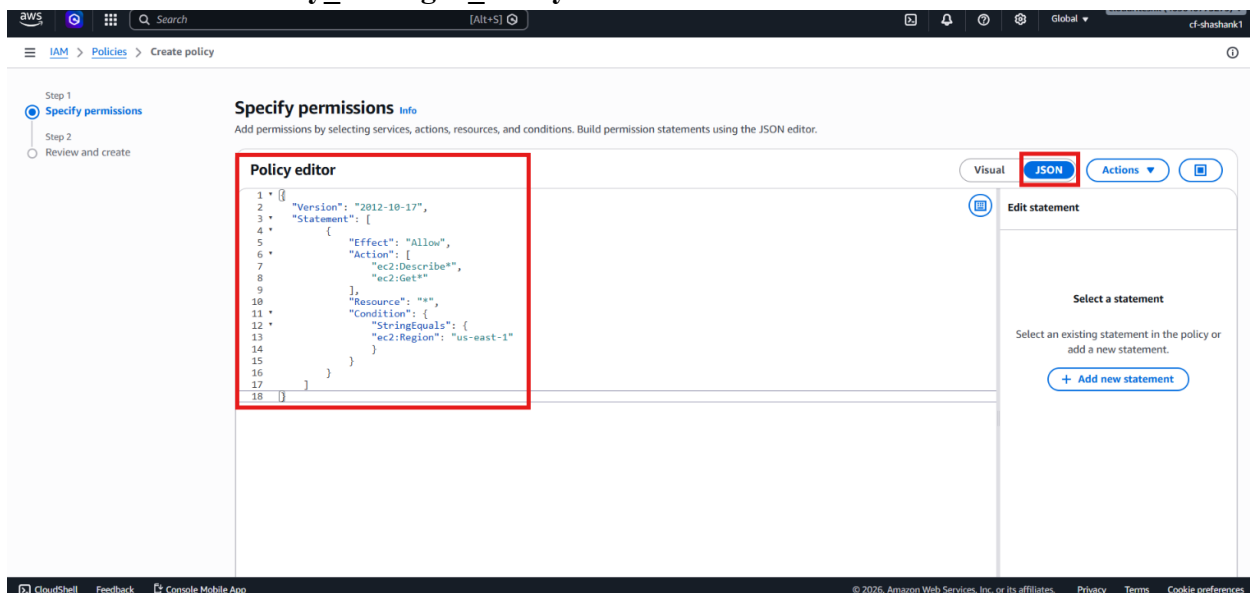
- Click **Create user**.
- Same steps for creating **user-2**.

Step 2 – Create a Managed Policy for Comparison

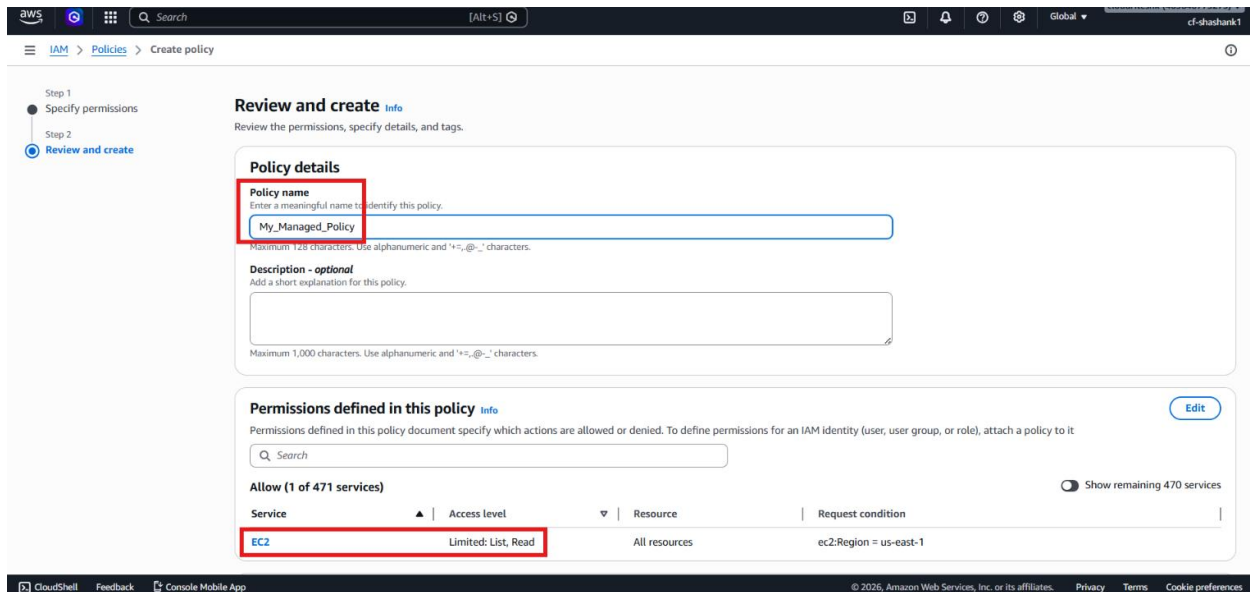
- Open **IAM** → **Policies**.
- Click **Create policy**.



- Choose the **JSON** tab.
- Paste the custom **My_Managed_Policy** from resources.

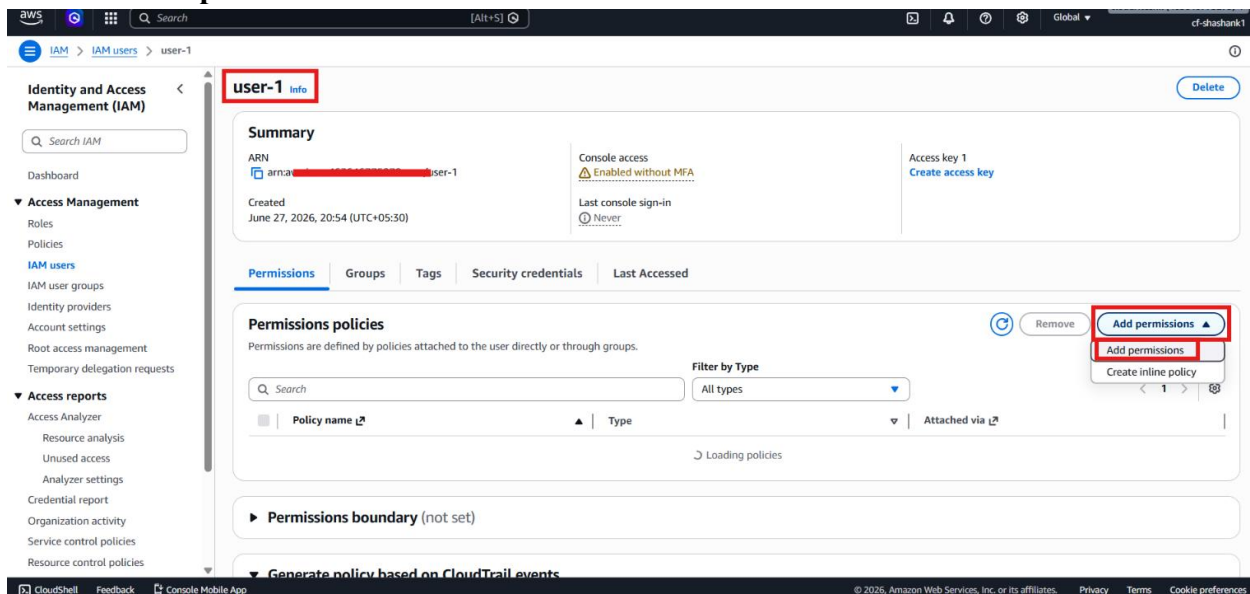


- This restrict access to **Amazon EC2** read-only operations in the selected Region.
- Name the policy: **MyManagedPolicy**
- Click **Create policy**.

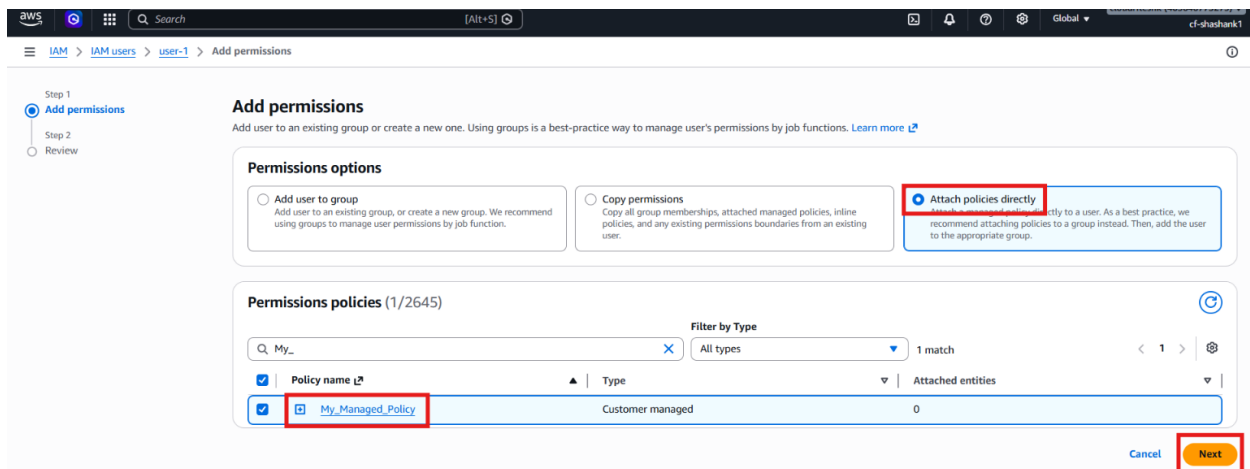


Step 3 – Attach the Managed Policy to Multiple Users

- Open user-1.
- Click Add permissions.



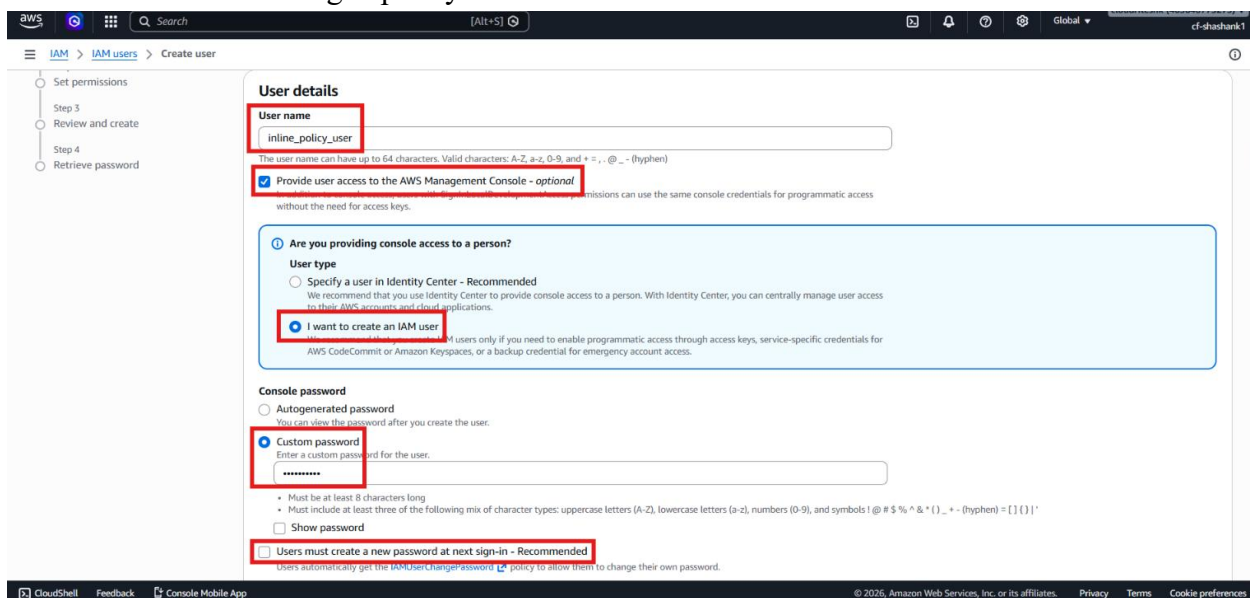
- Choose Attach policies directly.
- Select My_Managed_Policy.
- Click Add permissions.



- Open **user-2**.
- Repeat the same permission attachment steps.
- Confirm that the same managed policy can be reused across both users.

Step 4 – Create the Inline Policy User

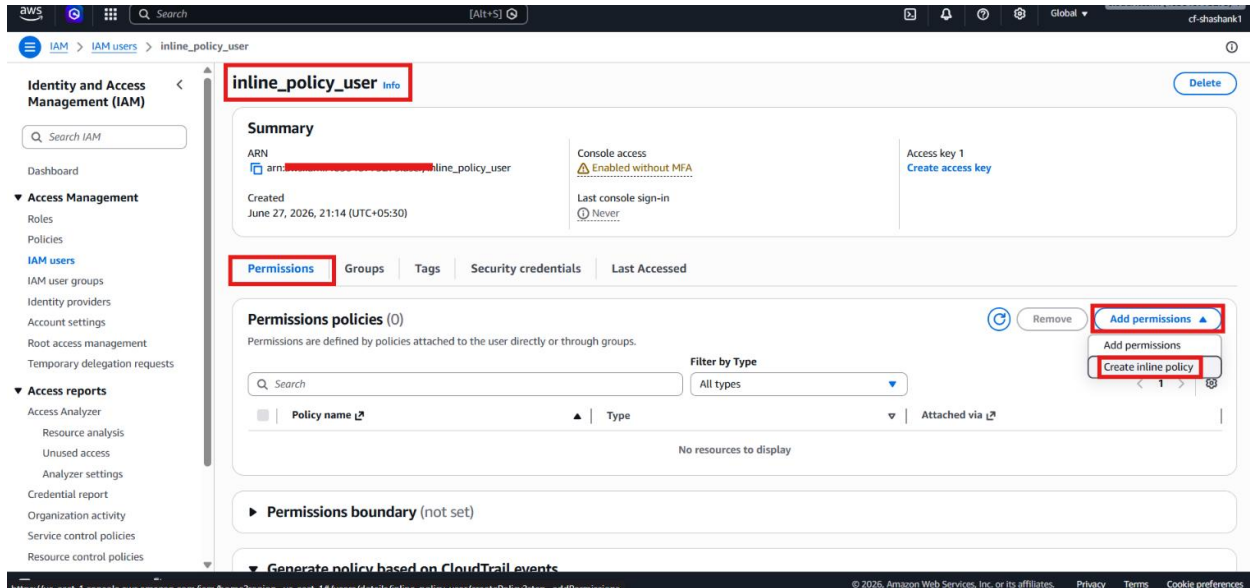
- Open **IAM** → **Users**.
- Click **Create user**.
- Create a user named **inline_policy_user**.
- Enable console access.
- Set a password.
- Do not attach a managed policy.



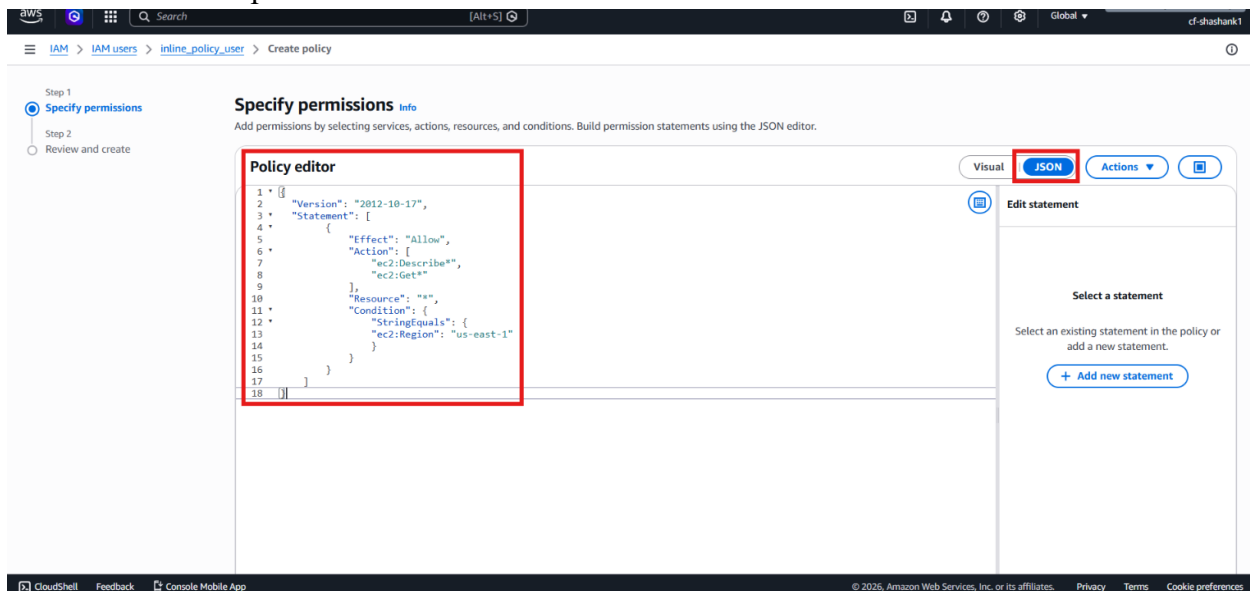
- Click **Create user**.

Step 5 – Create the Inline Policy

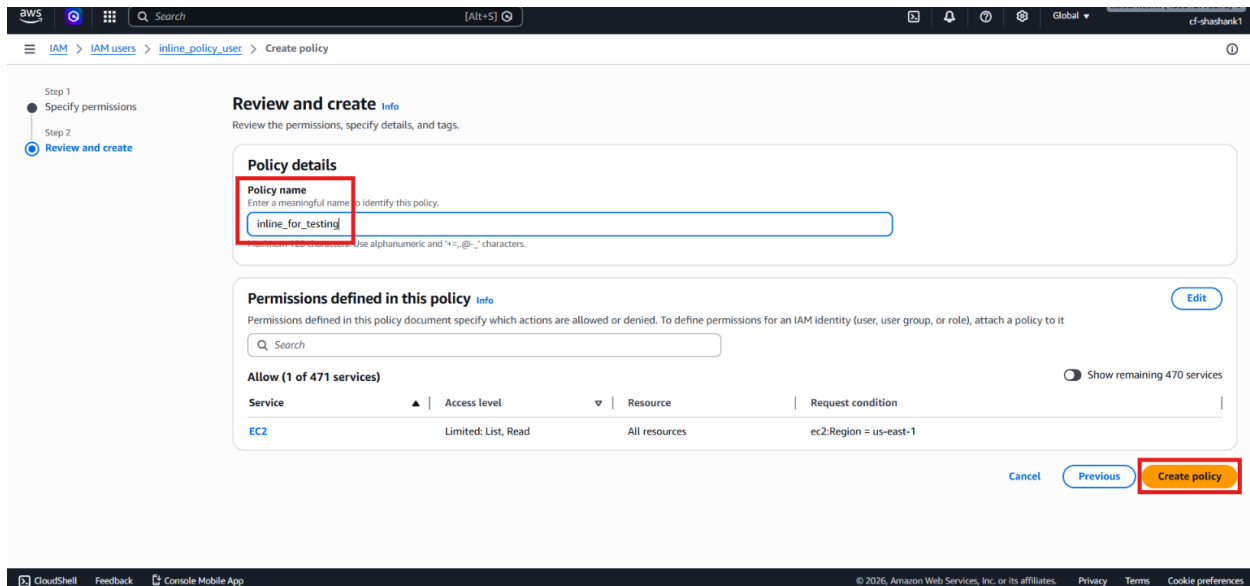
- Open `inline_policy_user`.
- Click **Add permissions**.
- Choose **Create inline policy**.



- Open the **JSON** tab.
- Paste the same permission document used earlier.

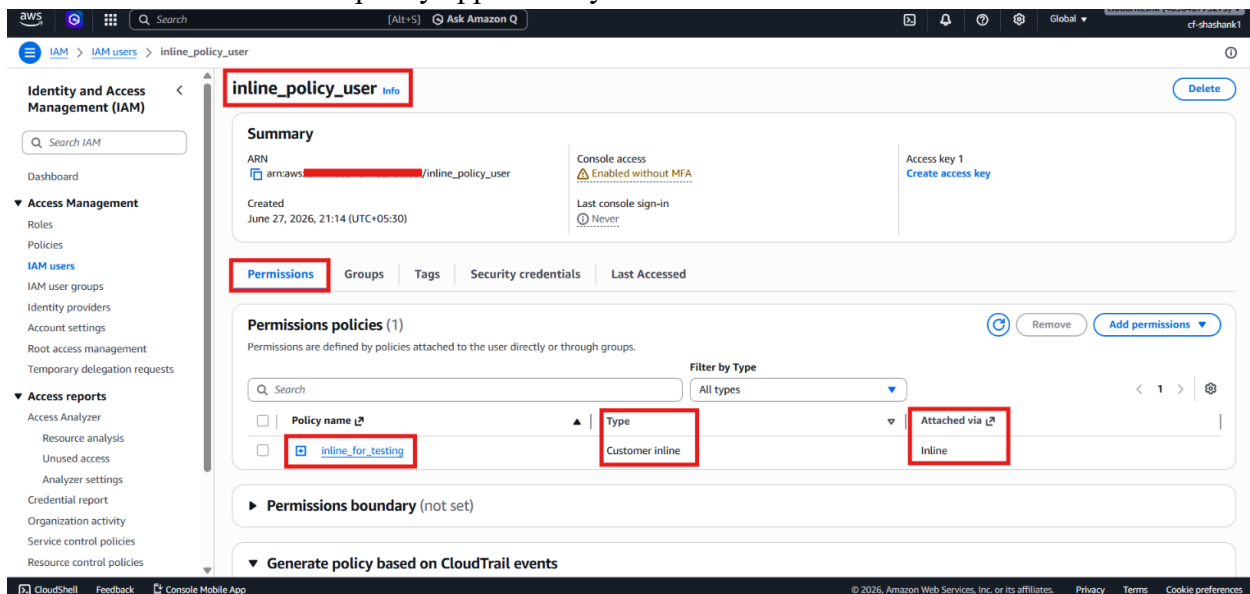


- Name the inline policy: `inline_for_testing`
- Create the policy.

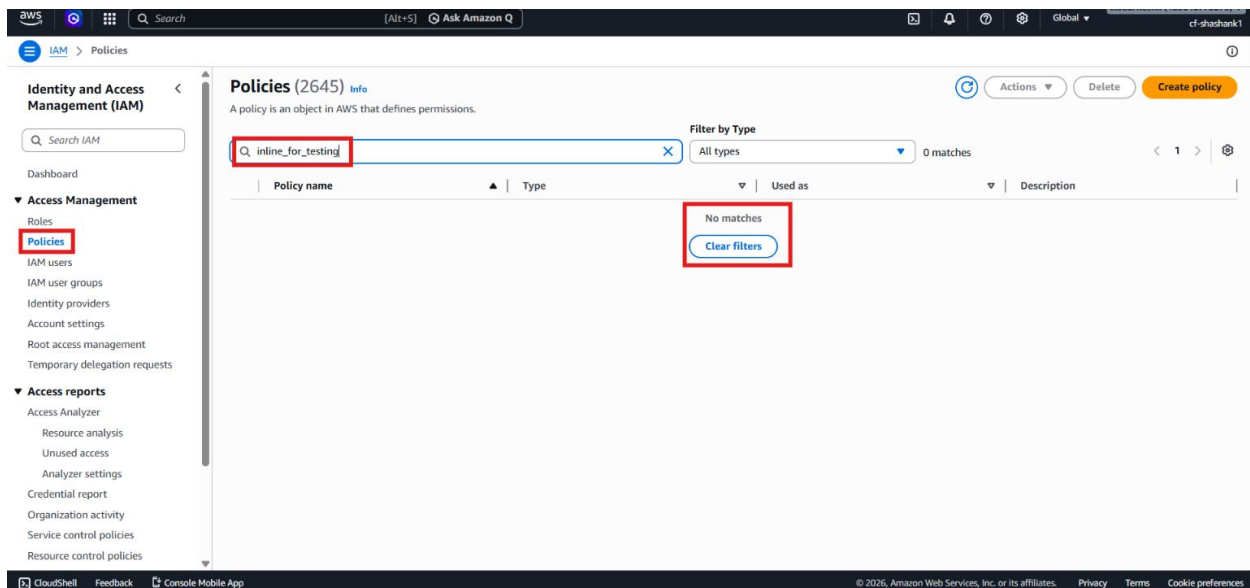


Step 6 – Verify Inline Policy Behavior

- Open the `inline_policy_user` permissions page.
- Confirm that the inline policy appears only under that user.



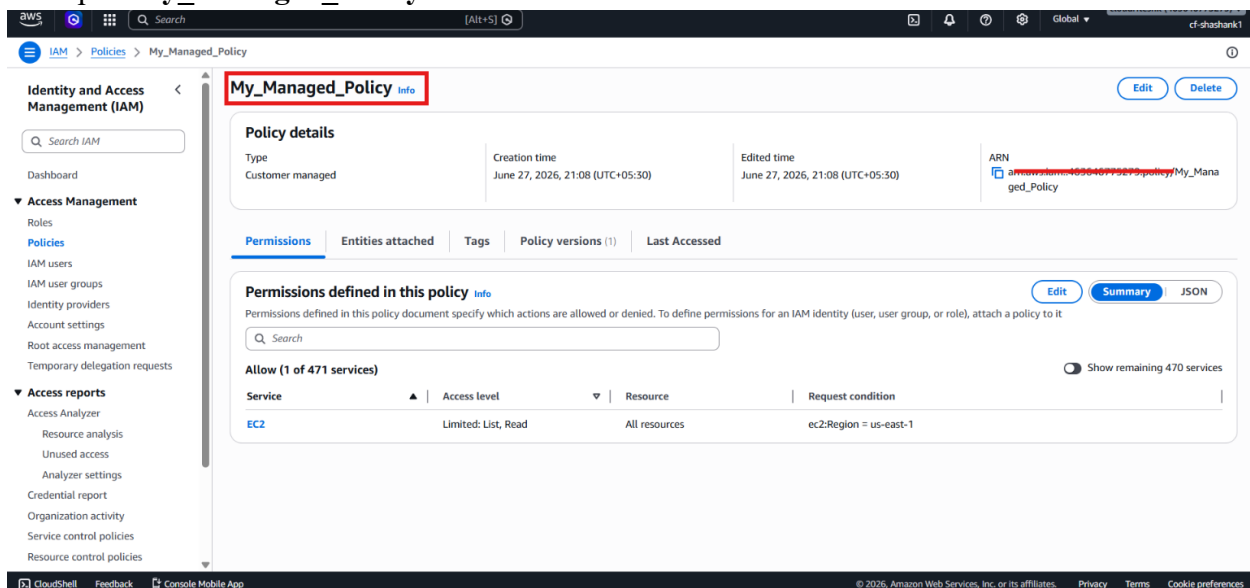
- Check the general policy list.
- Verify that the inline policy does not appear as a reusable standalone policy.



- Delete the IAM user.
- Confirm that the inline policy is deleted automatically with the user.

Step 7 – Compare Managed and Inline Policies

- Open My_Managed_Policy.



- Verify that the managed policy still exists even if **user-1** and **user-2** are removed.

Users deleted.

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Search

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☐	[redacted]	/	0	1 hour ago	-	18 days	1 hour ago	-

- Confirm that the managed policy can be attached to other users later.
- Compare this with the inline policy behavior.
- Confirm that inline policies are not reusable and are tied to only one entity.

Verification

- Two standard IAM users were created successfully.
- A reusable managed policy was attached to more than one user.
- An inline policy was created for a single IAM user only.
- The inline policy was not reusable across other entities.
- The inline policy was deleted automatically when its IAM user was deleted.
- The difference between managed policies and inline policies was verified successfully.